

6505 Lincoln Avenue
Carmichael, CA 95608
Phone: (916) 430-1628
Email: djmansperger@gmail.com
GitHub: github.com/djmans630

Daniel Mansperger

Fourth Year Data Science Student at UC San Diego

SKILLS

- **LANGUAGES:** Python, SQL, R, Java, HTML, CSS, JS, C, C++
- **LIBRARIES AND TOOLS:** Pandas, NumPy, SciPy, Matplotlib, Scikit-learn, MNE-Python, PyTorch, TensorFlow, Jupyter, Dask
- **MATHEMATICS:** Calculus, Linear Algebra, Differential Equations, Statistics
- **TECHNIQUES:** Feature Engineering, Statistical Modeling, Hypothesis Testing, Cross Validation, Model Evaluation, Time Series Analysis, Signal Processing, Machine Learning, Supervised Learning

EXPERIENCE

Premium Auditor-Cal Audits

June 2025 - Oct 2025, June 2024 - Sep 2024

Conducted remote audits of policyholder records and financial documents to verify insurance premiums in line with regulatory standards. Analyzed and validated structured financial datasets to identify discrepancies, ensure regulatory compliance, and maintain data integrity. Worked independently in a remote environment while managing deadlines across multiple audits.

Prior Audit Project-Cal Audits

Jul 2022 - Jul 2025

Managed high volumes of sensitive audit data with precision and consistency. Ensured data integrity and confidentiality while meeting strict deadlines. Developed strong independent work habits and time management skills.

Tutor-Sierra College Rocklin Campus

January 2023 - June 2023

PASS Peer and tutor selected by professor to support students in Differential Equations and Linear Algebra. Facilitated review sessions and helped clarify advanced math concepts, promoting student success and deeper understanding.

EDUCATION

Current Institution: University of California, San Diego

B.S. in Data Science, Expected June 2026

Transferred as a junior and currently developing skills in data analysis, statistical modeling, and computational methods. Coursework and projects emphasize data-driven problem-solving, effective communication, and technical rigor.

Previous Institution: Sierra College (Rocklin)

June 2021 – June 2023

Studied Engineering and Computer Science, building a strong analytical and technical foundation, with a particular emphasis on mathematics prior to transferring.

PROJECTS

Pre-screening motor imagery BCI literacy using low-sample EEG-derived features.

[Website](#) | [Repository \(Full\)](#) | [Repository \(Data Analysis\)](#) | [Research Study](#) | [Poster](#)

- Built an end to end machine learning pipeline to predict motor imagery BCI performance from low sample EEG data using classification and regression models
- Engineered 38 plus features including mu and beta band power, ERD and ERS dynamics, CSP based class separability, and signal stability; applied statistical feature selection using Spearman correlation with FDR correction
- Trained and evaluated regression and classification models including Random Forest, Gradient Boosting, SVM, and Ridge using cross validation, achieving Pearson r of 0.58 and 78.8 percent accuracy for high versus low performer classification
- Reduced required calibration data to approximately 15 trials, demonstrating a data efficient approach to early stage screening
- Developed reproducible data pipelines and modular code for feature extraction, model training, and evaluation using NumPy, SciPy, MNE, and scikit learn; built a web interface to visualize results
- Conducted exploratory data analysis and visualized EEG feature distributions to validate signal quality and feature relevance

Investigating the Relationship Between Minutes to Cook and Other Characteristics in a Recipe

[Website](#) | [Repository](#)

- Developed a regression model to predict cooking time using a dataset of 230,000 plus recipes, incorporating structured features and text-derived features from ingredients and instructions
- Performed extensive data cleaning and feature engineering on raw recipe data, transforming text and nutritional attributes into model ready features
- Conducted exploratory data analysis and hypothesis testing to identify relationships between recipe complexity and cook time
- Improved model performance through feature transformations and hyperparameter tuning with GridSearchCV, achieving consistent RMSE reductions
- Evaluated model fairness across recipe complexity groups using permutation testing

Making Healthier Choices

[Website](#) | [Repository](#)

- Collaborated on the design and development of an interactive web application to visualize macronutrient distributions and support healthier dietary decisions
- Implemented dynamic front end components using HTML, CSS, and JavaScript including responsive charts, dropdown filters, and interactive visualizations

- Integrated data driven insights into the UI to improve usability and user engagement while maintaining clean and accessible design